

RAJATH CHIKKAMAGALURU RAVI KUMAR

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SUMMARY

- Certified professional in RF Fundamentals, designing and measuring various RF/Microwave Circuits and Components.
 - Experienced in designing Low Noise Amplifier (LNA), indoor-DAS, Lumped and Distributed element Filters, Impedance Matching circuits, Couplers and other RF devices.
 - Hands-on experience in using RF lab equipment's such as Vector Network Analyser, Spectrum Analyser, Noise Meter etc., in measuring Mixers, Couplers, Amplifiers, Cushcraft and Patch Antenna.
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EDUCATION

Master of Science (MSEE), Electrical Engineering

MAY 2019

University of South Florida, Tampa, FL – 33647

GPA: 4.0/4.0

Relevant Coursework: Wireless Circuits/Systems, RF and Microwave design, RF and Microwave Measurements, Cellular Network and Wireless Network Architecture.

Bachelor of Engineering in Electronics and Communication Engineering

JULY 2017

Bangalore Institute of Technology, Bangalore, KA, India

CGPA: 8.75

Relevant coursework: Microwave and Radar, Antenna Propagation, Network analysis, Digital Logic Design, Control systems, Embedded System Design and Network Security.

TECHNICAL SKILLS

- **Simulation Software:** Keysight ADS, HFSS, AWR, MATLAB, C.
 - **RF/MW measurements:** S-Parameter measurements, Gain Compression, P1 dB measurement, Link Budget, indoor-DAS, IP3 or TOI measurement, PIM testing, On-wafer Probing, Phase Noise measurement, Noise Figure and Power measurement.
 - **RF Testing Equipment:** Vector Network Analyser (VNA), Spectrum Analyser, VCO, Power Meter, Synthesized Signal Generator, Noise meter, Soldering.
 - **CERTIFICATION:** *Keysight RF and Microwave Industry – Ready Student Certification.*
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PROJECTS

Low Noise Amplifier Design.

JAN 2019

- Designed, fabricated and measured a Low noise Amplifier using transistor “Infineon BFP420 NPN BJT” for assigned specifications at 1.4GHz using *Keysight’s ADS* and *Modelithics Library*.

Coupled Line Band Pass Filter Design.

NOV 2018

- Designed and fabricated a Coupled line Band Pass filter on FR4 substrate at 2.45GHz. EM simulation is performed and **Figure of Merit (FOM) of 16.02** was obtained from the measurement.

Distributed Microstrip Matching Circuit Design.

SEP 2018

- Designed a matching circuit to match a given load to 50ohms, obtained the maximum return loss bandwidth, performed Momentum simulation and created a layout for fabrication.

Design, characterization of Microstrip Patch Antenna.

MAR 2018

- Designed, fabricated a patch antenna and compared its return loss values with the values obtained for Cushcraft antenna.
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INDUSTRY EXPERIENCE

Independent Study, University of South Florida

JAN 2019 - Present

- The research was on designing an **indoor-Distributed Antenna System (DAS)** for improved cellular coverage and public safety applications.
- Issues regarding **Passive Intermodulation (PIM)**, **Received Signal Strength (RSSI)** and poor coverage were diagnosed and resolved.

Internship, Bharat Electronics Limited (BEL)

FEB 2017 - JUNE 2017

- Designed and Simulated a 9th order **Lumped Element Low Pass Filter** using **AWR Design** Software.
- Hands-on experience in calibrating and measuring the circuit parameters over Network Analyser.